

CSE0613211: Data Structures and Algorithms I

Part A: Course Information

1. **Program:** BSc. in CSE
2. **Course Code:** CSE0613211
3. **Course Title:** Data Structures and Algorithms I
4. **Course Type (GED/Core Course/Electives):** Core Course
5. **Year/Level/Semester/Term:** 2nd year/ 1st Semester
6. **Academic Session:** Spring 2026
7. **Section:** 3A
8. **Prerequisite Course (If any):** CSE211
9. **Credit Value:** 3.00
10. **Contact Hours:** 3.00 hours/Week
11. **Class Schedule:** Sunday (Room: 412, Period: 11.00 AM – 12.00 PM),
Monday (Room: 202, Period: 10.00 AM - 11.00 AM)
Wednesday(Room: 413, Period: 10.00 AM - 11.00 AM)
12. **Counseling Schedule:**
Sunday (12.00 PM – 02.00 PM),
Monday (8.00 AM – 10.00 AM)
Thursday (10:00 AM - 12:00 PM)
1. **Course Teacher:** Tahmina Islam, Designation: Lecturer, Room: 418(12), Email: tahmina.islam@uits.edu.bd,
Cell: +8801533534929, Specialization: Machine Learning, Deep Learning, Generative Adversarial Networks.
13. **Rationale of the Course:** This course introduces the structured way of handling data so that a complex problem can be solved efficiently. Students will learn to use and link data structures. Also, you will know something of all of these (sorting and searching, categorized efficiency in time and memory use, linked list and tree data structures, hash tables, stacks, and queues) by the end of the course.
14. **Course Objectives:**
 1. To introduce a structured way of handling data so that a complex problem can be solved efficiently.
 2. To use and design linked data structures.
 3. To know something of all of these (sorting and searching, categorized efficiency in time and memory use, linked list and tree data structures, hash tables, stacks, and queues) by the end of the course.
 4. To be familiar enough with the concepts that they should take any of them further and make use of them.

Part B: Skill Mapping

15. Course Outcome (COs)

No.	COs	BT	CP/WP	CA/EA	KP/WK	AT	DM&A
CO1	Explain the fundamental concepts of data structures and algorithms used in problem-solving and analyze their time	L2	–	-	-	CT, MT, FE	Lecture, Discussion

	and space complexity to evaluate algorithmic efficiency						
CO2	Demonstrate the working principles of linear and non-linear data structures and analyze algorithm efficiency	L4	–	–	K2, K3, K4	CT, MT, FE	Lecture, Discussion
CO3	Apply efficient data structures and algorithms to solve real-world problems.	L5	–	-	K5	FE, A	Lecture

*COs- Course Outcome; BT- Bloom's Taxonomy; CP/WP- Complex Problems; EA/WA- Complex Activities; AT- Assessment Tools; KP/WK- Knowledge Profile; DM&A: Delivery Methods & Activities; FE—Final Exam; CT—Class Test; MT—Midterm; A—Assignment.

16. Mapping of COs with Program Outcomes (POs)

Course Outcomes (CO)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2		√										
CO3			√									

Part C: Teaching Learning Approach

17. Course plan specifying contents, COs, co-curricular activities (if any), teaching learning and assessment strategy mapped with COs.

*Commencement of the semester (Autumn 2025): 13 July 2025, and Last class of the semester: 20 November 2025

Week	Topics	Suggested Activity and Teaching Strategy	Assessment Strategy	Corresponding COs
1	Data Structure, Classification of data structure, Operations on data structure, Algorithm and Program; Complexity of an algorithm: Time complexity, Space complexity, Time space tradeoff, mathematical notation, and function.	This activity aims to introduce students to the core concepts of data structures and algorithms , exploring their real-world applications, performance analysis, and connections to neural networks. Students will engage in discussions and interactive demonstrations, facilitating a deeper understanding of these essential computing concepts.	Class Test 1, Midterm	CO1

2	Introduction to Data Structures and Linear Arrays: Importance of data structures, mathematical and algorithmic notations, control structures, asymptotic analysis (Big O, Ω, Θ), linear arrays, and basic array operations (searching, insertion, deletion, and summation).	Students will develop a strong foundation in data structures, algorithmic thinking, and complexity analysis through lectures, worked examples, step-by-step algorithm tracing, real-world problem-solving activities, and hands-on programming exercises involving linear array operations.	Class Test, Midterm	CO1, CO2
3	Two-Dimensional Array: definition, graphical representation, C/C++ representation, and memory representation. Two-Dimensional Array: location of an element of a two-dimensional array, algorithm to find out the summation of boundary elements and diagonal elements.	Students will learn how to work with two-dimensional arrays by exploring their definition, graphical and memory representations, and C/C++ syntax, while also understanding how to locate elements within the array and applying algorithms to find the summation of boundary and diagonal elements through conceptual examples.	Class Test, Midterm	CO1, CO2
4	Searching: definition, linear and binary search; Sorting: definition, selection, and insertion sort.	Students will study searching algorithms, including linear and binary search, and sorting algorithms, such as selection and insertion sort. In Class Test 1, they will analyze the efficiency of these algorithms, evaluate their performance, and optimize their application to various problem-solving scenarios.	Class test 1, Midterm	CO1, CO2, CO3
5	Linked List: definition, types; Linear Linked List: definition, algorithm to create a linked list; Algorithm on Linear Linked List: algorithm to search a node from a linked list, algorithm to insert a node into a linked list, algorithm to delete a particular node from a linked list, differences between array and linked list. Doubly and Circular Linked List	Students will explore the fundamentals of linked lists, including their definition and various types such as Linear Linked Lists , Doubly Linked Lists , and Circular Linked Lists . They will study algorithms to create a linked list, search for a node, insert a node, and delete a particular node. Additionally, students will compare the differences between arrays and linked lists in terms of memory management and data handling.	MidTerm Exam	CO1, CO2

6	Sorting: bubble sort, merge sort, and quicksort	Students will examine various sorting algorithms, including bubble sort, merge sort, and quicksort, understanding their unique characteristics, time complexities, and use cases.	Assignment, Final Exam	CO1, CO2, CO3
7	Stack: definition, array and linked-based stack, PUSH and POP operation; Applications of Stack: checking the validity of an arithmetic expression, converting an infix arithmetic expression to its postfix expression, evaluating a postfix expression; Tower of Hanoi problem;	Students will explore the foundations of stacks by understanding their definition and the differences between array-based and linked-based stacks. They will further study the PUSH and POP operations through practical examples and discuss how stacks are used in real-world applications such as checking the validity of an arithmetic expression, converting an infix expression to postfix, evaluating a postfix expression, and solving the Tower of Hanoi problem. Through these examples, students will grasp the significance of stacks in managing data in a Last In, First Out (LIFO) manner and its role in algorithmic problem solving.	, Final Exam	CO1, CO2
8	Queue: definition, array and linked-based queue, algorithm to add an element to queue, algorithm to delete an element from a queue; Priority Queue. Implementation of a stack using a queue and implementation of a queue using a stack;	Students will explore the fundamentals of queues, including their definition, and the differences between array-based and linked-based queues. They will study algorithms to add and delete elements from a queue, as well as the concept of a Priority Queue . Additionally, students will learn how to implement a stack using a queue and a queue using a stack. Through these exercises, students will understand the underlying principles of FIFO (First In, First Out) data management and its practical applications in real-world scenarios.	Final Exam	CO1, CO2
9	Graph: Basics of graph, adjacency matrix, and list of a graph; Graph Traversal Methods: BFS and DFS	Students will explore the fundamentals of graphs, including their basic structure and how graphs are represented using adjacency matrices and adjacency lists . They will study graph traversal methods such as Breadth-First Search (BFS) and Depth-First Search (DFS) , understanding how these algorithms systematically explore nodes and edges in a graph. Through these concepts, students will learn how graphs are utilized in various applications such as network routing, social networks, and pathfinding algorithms.	Assignment, Final Exam	CO1, CO2, CO3
10	Tree: definition, binary tree, tree implementation, full and complete binary tree, parent-child relationships; Tree Traversal Techniques: Pre-order, In-order, and Post-order.	Students will explore the fundamentals of trees, including their definition and key concepts such as binary trees , full and complete binary trees , and the parent-child relationships that structure them. They will study the implementation of binary trees and learn the differences between various types of binary trees. Additionally, students will focus on tree traversal techniques , including Pre-order , In-order , and Post-order traversal, understanding how each method visits the nodes in a specific order. These traversal techniques are essential for tasks such as searching, sorting, and processing hierarchical data in applications like expression evaluation and decision-making processes.	Assignment, Final Exam	CO1, CO2, CO3

11	BST: definition, creation procedures; Heap: definition, types, creation procedures; Huffman coding	Students will explore Binary Search Trees (BSTs) , understanding their definition and the procedures for creating a BST. Students will also delve into Heaps , studying their definition, types (such as Max Heap and Min Heap), and creation procedures. Heaps are essential for priority queue implementations and efficient sorting algorithms. Finally, students will learn about Huffman Coding , a lossless data compression algorithm used to minimize the size of files by encoding data using variable-length codes based on frequency.	Assignment, Final Exam	CO2, CO3
12	Minimum Cost Spanning Tree: Spanning subgraph, spanning tree, minimum cost spanning tree, Prim's and Kruskal's algorithm, differences between tree and graph.	Students will explore the concept of Minimum Cost Spanning Trees (MST) , starting with the definitions of spanning subgraphs and spanning trees . They will learn how a spanning tree is a subgraph that connects all the vertices of a graph without any cycles and includes all the edges required to connect those vertices. The focus will then shift to Minimum Cost Spanning Trees , where the goal is to find a spanning tree with the least possible total edge weight. Students will study two well-known algorithms for finding the MST: Prim's Algorithm and Kruskal's Algorithm .	Assignment, Final Exam	CO2, CO3
13	Hashing: introduction, Hash function, Methods of hashing: division, mid-square, folding, Hash collision resolution scheme: linear and quadratic probing, random and double hashing, rehashing, chaining	Students will explore hashing , including the hash function and methods like division , mid-square , and folding , and learn about collision resolution schemes such as linear probing , quadratic probing , random hashing , double hashing , rehashing , and chaining to efficiently store and retrieve data in hash tables.	Assignment, Final Exam	CO1, CO2, CO3
14	Term Final Suggestion and Problem Analysis	Answer any type of question from students and solve their problems.	Final Exam	

Part D: Assessment Approach

18. Assessment and Evaluation

Assessment Components		Marks Distribution
Continuous Internal Assessment (CIA) (50%)	Class Attendance/Class Performance	10%
	Class Test/Assignment/Quiz/Surprise Test	20%
	Midterm	20%
Final Exam (FE)		50%
Total		100%

19. Assessment Pattern

1. Continuous Internal Evaluation (50 Marks)

*The percentage distribution of Bloom's categories in the assessment tools may vary by $\pm 5\%$.

Bloom's Category	Class Test 1/ Assignments 1	Mid Term Exam	Class Test 2/ Assignments	Attendance & Class Participation
Marks	(10)	(20)	(10)	(10)
Remember		2%		
Understand	5%	6%		
Apply	5%	8%	5%	
Analyze		4%		
Evaluate				
Create			5%	

2. Final Exam Evaluation (50 Marks)

*The percentage distribution of Bloom's categories in the assessment tools may vary by $\pm 5\%$.

Bloom's Category	Final Term (50)
Remember	
Understand	10%
Apply	20%
Analyze	10%
Evaluate	5%
Create	5%

20. Grading system

Numerical Grade	Letter Grade	Grade Point
80% and above	A + (A plus)	4.00
75% to less than 80%	A (A regular)	3.75
70% to less than 75%	A – (A minus)	3.50
65% to less than 70%	B + (B plus)	3.25
60% to less than 65%	B (B regular)	3.00
55% to less than 60%	B – (B minus)	2.75
50% to less than 55%	C + (C plus)	2.50
45% to less than 50%	C (C regular)	2.25
40% to less than 45%	D	2.00
Less than 40%	F	0.00

Part E: References

21. Learning Materials

1) Recommended Readings:

a) Schaum's Outline of Theory and Problems of Data Structures, Latest Edition, Seymour Lipschutz.

2) Supplementary Readings:

a) Mark Allen Weiss, Data Structures and Algorithm Analysis in C++, Third Edition, Addison Wesley.

b) Data Structure Fundamentals-Md. Rafiqul Islam, M. A. Mottalib

3) **Others** (as applicable for the discipline academic program): Provided materials in google classroom.

Part F: Additional Information

- i. University of Information Technology and Sciences. Program outcomes. <https://uits.edu.bd/departement-of-computer-science-engineering/>
- ii. University of Information Technology and Sciences. Outcome-based curriculum. <https://uits.edu.bd/curriculum-of-cse/>
- iii. University of Information Technology and Sciences. Course catalog of Computer Science and Engineering. <https://uits.edu.bd/course-catalog-of-computer-science-engineering/>
- iv. University of Information Technology and Sciences. (n.d.). Faculty members of the Department of Computer Science and Engineering. <https://uits.edu.bd/faculty-members-of-cse/>
- v. University of Information Technology and Sciences. (n.d.). Bloom's taxonomy, knowledge profile, complex engineering problems, and complex engineering activities.
- vi. University of Information Technology and Sciences. Make-up examination procedure.
- vii. University of Information Technology and Sciences.. Attendance policy.
- viii. University of Information Technology and Sciences. Academic calendar. <https://uits.edu.bd/academic-calendar/>
- ix. University of Information Technology and Sciences. Academic policies.
- x. University of Information Technology and Sciences. (2024). Code of conduct. <https://uits.edu.bd/wp-content/uploads/2024/01/Code-of-Conduct.pdf>